

Creating specialized classes for fire risk using machine learning methods.

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Abstract

This study explores various machine learning techniques for assessing forest fire risk. The data used in this study have been selected from three different areas in Greece over the past 25 years (2000 - 2024). The first area of interest is Attica, the second area is eastern Macedonia and Thrace, and the third area is Crete. For each of the aforementioned areas, a separate data file was created, an analysis was carried out, and specialized definitions - classes were created for the forest fire risk occurrence. This study utilized a range of traditional machine learning techniques alongside novel methods based on Grammatical Evolution, including automated feature construction from existing attributes and the automatic generation of classification rules.

Keywords: Fire risk; Machine learning; neural networks; Grammatical evolution.

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4.1.1. Subsubsection

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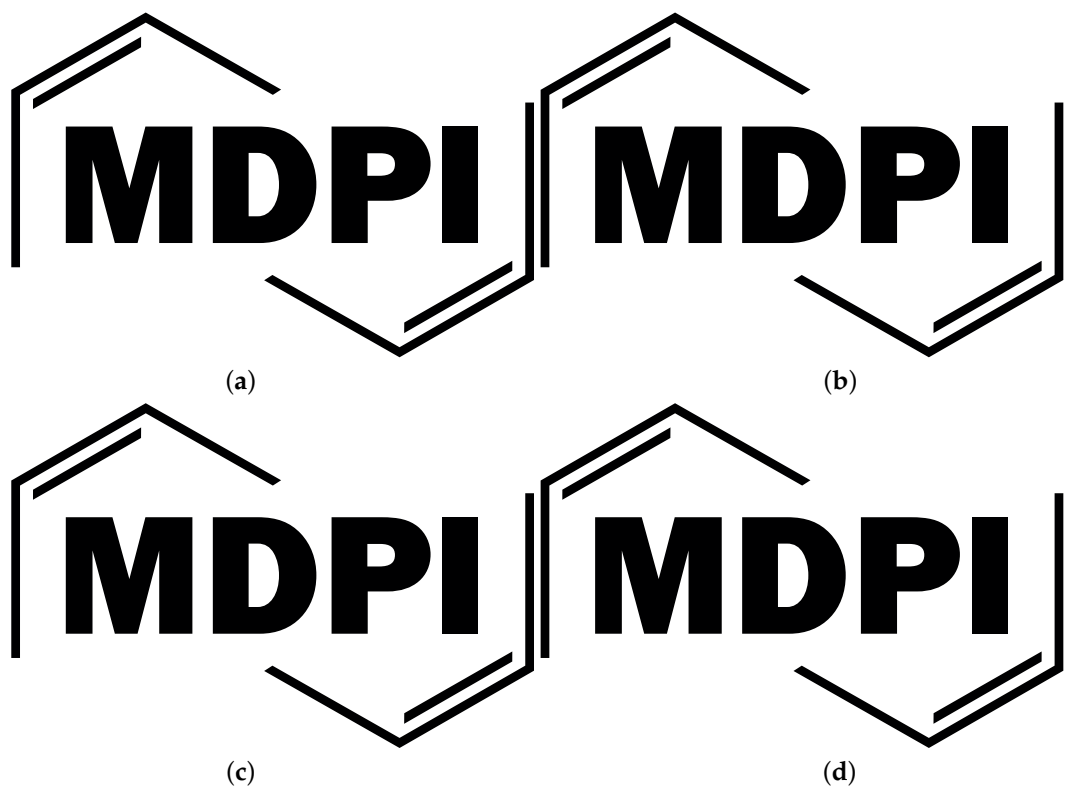


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	Data	Data	Data
	Data	Data	Data
Entry 2	Data	Data	Data
	Data	Data	Data
	Data	Data	Data

* Tables may have a footer.

Text.

Text.

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76

This is the example 1 of equation:

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$$a = 1,$$

(1)

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the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

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This is the example 2 of equation:

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$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z$$

(2)

82

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Abbreviations

The following abbreviations are used in this manuscript:

- MDPI Multidisciplinary Digital Publishing Institute
- DOAJ Directory of open access journals
- TLA Three letter acronym
- LD Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data are shown in the main text can be added here if brief, or as Supplementary Data. Mathematical proofs of results not central to the paper can be added as an appendix.

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Entry 2	Data	Data

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References

1. Author 1, T. The title of the cited article. *Journal Abbreviation* **2008**, *10*, 142–149.

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2. Author 2, L. The title of the cited contribution. In *The Book Title*; Editor1, F., Editor2, A., Eds.; Publishing House: City, Country, 2007; pp. 32–58.

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4. Author 1, A.B.; Author 2, C. Title of Unpublished Work. *Abbreviated Journal Name* year, phrase indicating stage of publication (submitted; accepted; in press).

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5. Title of Site. Available online: URL (accessed on Day Month Year).

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